written; see Manjit Kumar 2008, *Quantum: Einstein, Bohr, and the Great Debate About the Nature of Reality* (Icon Books/Norton); or David Lindley 2007, *Uncertainty: Einstein, Heisenberg, Bohr, and the Struggle for the Soul of Science* (Anchor) for good examples.

**"countless arithmetical errors":** Heisenberg 1971, p. 61.

**"so generously spread out before me":** Ibid.

**"generally my severest critic":** Ibid., p. 64.

**"once again possible to move forward":** Kumar 2008, p. 193.

**"purely a thought-thing":** Isaacson 2007, p. 84.

**"profound influence on me":** Albert Einstein 1949a, "Autobiographical Notes," in *Albert Einstein: Philosopher-Scientist,* edited by Paul Arthur Schilpp (MJF Books, 1949), p. 21.

**he was not a dogmatic Machian:** See Don Howard 2015, "Einstein's Philosophy of Science," in *The Stanford Encyclopedia of Philosophy,* Winter ed., edited by Edward N. Zalta, http://plato.stanford.edu/archives/win2015/entries/einstein-philscience/. Also see Chapter 8 for more on the influence Einstein had among Mach's followers and their reactions upon learning his true philosophical views.

**introduced by the physicist Max Planck:** Gerald Holton 1998, *Thematic Origins of Scientific Thought,* rev. ed. (Harvard University Press), p. 70.

**"invariant theory":** Ibid., p. 130.

**"essentially untenable":** Einstein 1949a, p. 21.

**"It can only exterminate harmful vermin":** Isaacson 2007, p. 334.

**"determining what *is*":** Kumar 2008, p. 262. Emphasis in original.

**a few holdouts:** The last holdouts were finally forced to accept the reality of photons as a side effect of the Bell experiments conducted by John Clauser in the 1970s. See Chapter 9.

**"one-horse cart":** Kumar 2008, p. 35.

**prejudices accumulated by the age of eighteen:** Lincoln Barnett 1949, *The Universe and Dr. Einstein* (Victor Gollancz), p. 49.

**"In Göttingen they believe in it. I don't":** Isaacson 2007, p. 331.

**"such strange assumptions":** Heisenberg 1971, p. 62.

**"restrict myself to these":** This was likely a post hoc justification Heisenberg gave for his own work. His true motivation in ignoring orbits was probably that they had proven largely useless over the previous decade in explaining new experimental results. See Beller 1999b,